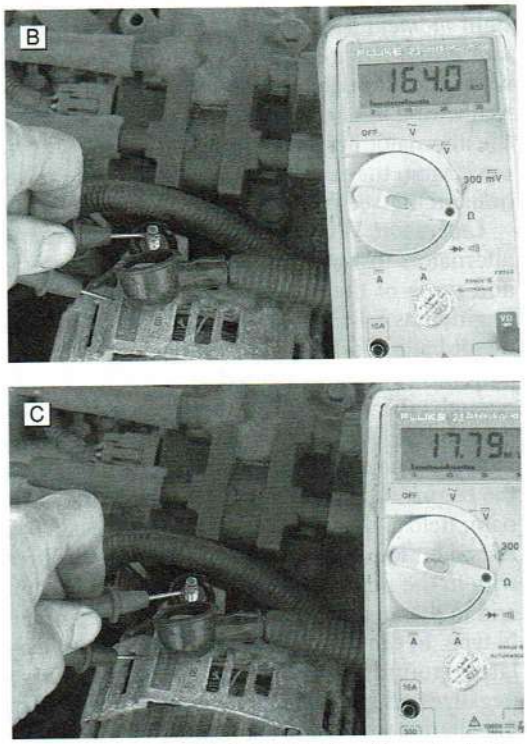
Troubleshooting Chart 3-3.
Battery/Alternator/Regulator Problems: No Alternator Output

**Symptoms:** Ignition warning light stays on, battery remains discharged.

| | |
|---|---|
| Is the drive belt loose or broken?<br>**NO** **TEST:** The belt should respond to finger pressure with no more than ⅜ to ½ inch of give at its midpoint. | **YES** **FIX:** Tighten or replace. |
| Is the circuit between the alternator and battery at fault?<br>**NO** **TEST:** Either (1) test for output with a clamp-on DC amp meter, or (2) shut down the engine, isolate or shut down auxiliary chargers (solar panels, wind generator, etc.), turn on battery isolation switch, turn on engine ignition switch (but do not start engine), and check voltage (a) across the battery terminals and (b) between alternator output and ground. If voltage at alternator output terminal is neither about the same as the battery voltage (no isolating diodes) or 0 (isolating diodes, dual battery installation), the circuit between alternator and battery is faulty. (Note: Diodes may sometimes show voltage—see text.) | **YES** **FIX:** Check all wiring and connections. Clean and replace as necessary. |
| Does the alternator have an internal voltage regulator?<br>**NO** **TEST:** Look for absence of alternator field terminal, field wire, and external regulator. | **YES** **FIX:** If a lack-of-output problem has been traced this far and the regulator is internal, you have little choice but to install a new alternator. |
| If the voltage regulator is external, is it a P-type or an N-type?<br>**TEST:** Shut down engine, turn off ignition and battery isolation switches, and disconnect field wire from alternator. Measure resistance between field terminal and ground: P-type = reading near 0 ohms; N-type = high reading. | **YES** **P-type FIX:** Proceed with Steps 1–3 below.<br>**YES** **N-type FIX:** Proceed with Steps 4–6 below. |
| (1) Does the alternator fail to work when the P-type voltage regulator is bypassed?<br>**NO** **TEST:** Connect a test lamp between alternator field terminal and battery positive terminal. Start engine. Look for rising battery voltage. | **YES** **FIX:** Alternator is broken. Check for damaged brushes and slip rings, or shorted or open-circuited rotor, stator, or diode. See pages 99–104 for possible repair procedure.<br>*continued* |

reading any leakage current through the alternator's diodes. This will generally be on the order of 0.5 milliamp, but in any event *should not be higher than 2.0 milliamps.* If it is, one or more of the diodes has a problem.

3. *With the engine shut down, disconnect all cables from the alternator output stud.* Test with an ohmmeter from the alternator output stud to ground, and then reverse the meter leads and test again (Figures 3-36B and 3-36C). In theory, the meter should show an infinite (very high) resistance in one direction and a low resistance (down toward 0 ohms) in the other direction. However, things are not this simple, because it requires a certain "forward voltage" to "open" *(bias)* the diodes in their conductive direction, and this has to be generated by the meter's internal battery. Depending on the meter and the state of its battery, we may get anything from infinity in both directions (most likely), to infinity (or a very high resistance) in one direction and a relatively low resistance in the other direction. If the meter has a specific *diode-testing* capa-




**FIGURES 3-36B AND 3-36C.** Testing alternator diodes with all leads disconnected from the alternator output (positive) stud. Using the ohms position on the meter, there should be a very high resistance in one direction and a low resistance with the meter leads reversed. But most meters will show a high reading in one direction (in this case 164 kiloohms) and a very high reading in the other direction (in this case 17.79 megaohms).